PCA-Based Shape Analysis of Abdominal Aortic Aneurysm Progression

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Introduction

Abdominal aortic aneurysm (AAA) surveillance is still primarily based on maximum diameter, despite the well-known heterogeneity of aneurysm growth and clinical behaviour. Aneurysms with similar diameters may exhibit markedly different growth patterns and outcomes, suggesting that diameter alone does not reliably capture disease progression [1,2].
In this study, we used a multivariate analysis of longitudinal CT-derived geometric descriptors to identify common morphological progression patterns and to investigate whether principal component analysis (PCA) [3] can reveal a coherent direction of aneurysm evolution and the descriptors that drive it.

Methods

A cohort of 123 AAA patients from the Karolinska Institutet (Stockholm, Sweden) was analysed, including patients' follow-up CT examinations for longitudinal assessment. Segmentations were standardized by restricting the region of interest to the infrarenal abdominal aorta, bounded superiorly by the renal arteries and inferiorly by the iliac bifurcation (Figure 1).

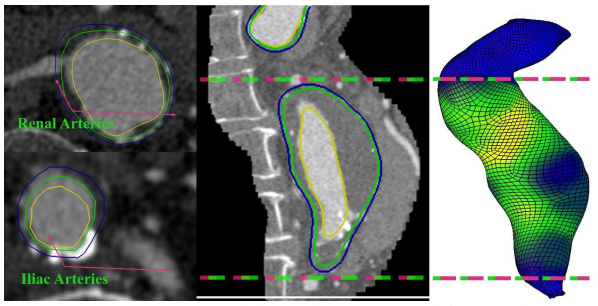


Figure 1: Representative abdominal CT scan used for morphological analysis. The anatomical truncation planes at the level of the renal arteries and the iliac bifurcation are indicated by coloured dashed lines.

From each three-dimensional segmentation, principal shape eigenvalues (λ_1, λ_2, λ_3), elongation, flatness, AAA volume, and a normalized shape complexity index were extracted. PCA was computed using baseline examinations only, defining a fixed morphological reference space. The last available CT of each patient was then projected onto the same space to visualize longitudinal trajectories and quantify displacement along the principal components.

Results

Baseline PCA identified a dominant component (PC1) mainly associated with aneurysm volume and shape complexity. Projection of follow-up examinations into the baseline space revealed a consistent rightward displacement along PC1 in nearly all patients (Figure 2). This shift was highly significant ($p < 0.001$), indicating a shared direction of shape evolution across the cohort.
The contribution of the descriptors changed markedly over time: volume became the dominant contributor to the PCA variance structure (0.79 to 1.88), whereas shape complexity showed a substantial reduction in importance (0.79 to 0.35). This temporal reweighting suggests that early AAA morphology is influenced by both size and geometric irregularity, while later stages are increasingly dominated by volumetric remodelling.

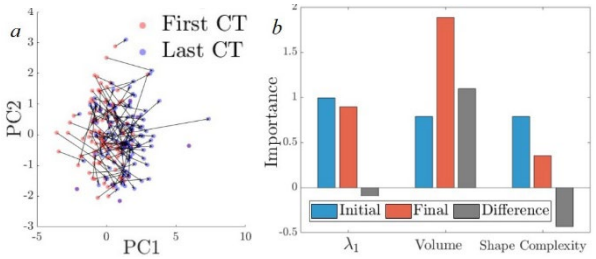


Figure 2: a) Baseline PCA reference space showing baseline examinations (red), follow-up examinations (blue), and patient-specific displacement vectors (arrows). b) Relative importance of the principal eigenvalue (λ_1), volume, and shape complexity in the baseline and follow-up PCA models.

Conclusions

This study highlights a consistent displacement of follow-up AAAs within a baseline-defined PCA space, suggesting a shared trajectory of aneurysm remodelling across patients. The increasing contribution of volume and the decreasing contribution of shape complexity indicate a possible temporal reorganization of AAA morphology during disease evolution, supporting the potential value of PCA-based geometric analysis beyond diameter alone.

Acknowledgements

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